



HACKOWATT / SCENARIO 2 / TECHNICAL EVIDENCE

Renewable Energy Simulator Handbook

A traceable explanation of the product, equations, assumptions, optimisation, thermal models, forecast evidence and presentation story.

4 kWp example size	4,000 kWh solar / year	9.4 y payback A	8.8 y payback B
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Reference example: generated 2024-2025 Silesian household, organiser economics, 1,000 kWh/kWp/year. Values are scenario estimates, not measured field performance.

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INSIDE THIS GUIDE

Product story

Challenge fit and user value

Models and formulas

Sources, constants, limits

Demo and Q&A

5-minute and 8+2 scripts

How to use this handbook

Use the first pages to explain the product and challenge fit. Keep the worked result, constants and jury questions open during the demo. The later sections distinguish implemented behaviour from the production roadmap.

USER VALUE	TECHNICAL EVIDENCE	HONEST LIMITS
One decision per screen; reversible recommendations; accessible controls.	Energy conservation, constrained schedules, thermal state and chronological forecasts.	Synthetic home, high current forecast error, static snapshots and simplified billing.

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1 - What the product does

A family chooses the size of a rooftop solar installation, sees what it would change, and decides which activities may move. The application shows the effect on their bill, electricity bought from the grid and their daily routine. A separate temperature simulation explores electric heating and hot water.

Challenge requirement	Where to demonstrate it
Select installation capacity in kWp	Shared solar slider, 0-10 kWp in 0.5 kWp steps
Annual production	Solar investment, metric 01
Share of household demand covered	Metric 02: self-used solar / total demand
Reduction of electricity purchased	Metric 03, relative to the original home without solar
Savings	Metric 04, after annual maintenance
Payback A: existing habits; B: activity shifting	Side-by-side payback cards and capacity comparison
Several capacity options	0, 2, 4, 6, 8 and 10 kWp, each with production, savings and payback
Predict, explain, optimise	Energy plan: forecast, historical replay, explanations and editable permissions

The team planning document informed the journey: forecast → explain peaks → suggest understandable changes → compare an investment. The application separates a replay, which knows the past, from a forecast, which must use only information available at its issue time.

The prototype is an offline advisory application. It does not switch real devices, poll weather services or settle an actual electricity bill. Synthetic household history is not a measured Polish family's consumption. These distinctions are visible in the interface.

2 - Data and provenance

Input	Purpose	What it represents
hourly.csv	Demand, weather, presence, device categories	Generated household on the supplied weather timeline
appliance_events.csv	Individual device sessions, minutes, energy and resource conflicts	Synthetic events from behaviour and device rules
residents_hourly.csv	Individual activity and presence audit	Person-hours, not physical appliance operating hours
resolved_house.json	House and tank properties	One resolved draw from the configured distributions; not inferred from a meter
run_config.json	Seed, behaviour and thermal assumptions	Reproducibility record
forecast_predictions.csv, forecast_metrics.csv	Prediction curves, error ranges, model comparison	Saved rolling-origin validation results
Organiser PDFs	Scenario and common financial assumptions	Authoritative for Hackathon mode
Optional team CSV and sensor JSON	Replace a load history or issued forecast	User-provided snapshot; provenance and freshness still need validation

The current full history contains **17,544 physical hours across 2024-2025**. UTC timestamps are the unique join key; local tariffs and routine displays use **Europe/Warsaw**. A daylight-saving day can contain 23 or 25 physical hours. Repeated local hours are distinct in UTC; the table includes the time-zone abbreviation.

Source wind is km/h, converted to m/s inside the pipeline by dividing by 3.6. Never pass km/h into the thermal model's **wind_ms** field. Weather is historical input; the irradiance column is an instantaneous horizontal-radiation sample, not a roof-plane energy measurement.

Annual averages currently divide by the number of calendar years represented. Use complete years for investment claims. A 30-day household sample may satisfy the challenge's minimum history requirement, but it does not establish annual heating demand; extrapolation would need a separate seasonally validated model.

3 - Units without jargon

Term	Plain meaning	Example
PV	Photovoltaics: sunlight converted into electricity	Rooftop solar panels
kW	Power: how fast energy is being used or produced	A 2 kW kettle
kWh	Energy: power integrated over time	$2 \text{ kW} \times 5/60 \text{ h} = 0.167 \text{ kWh}$
kWp	Rated peak panel power under standard test conditions	4 kWp is a size, not 4 kWh every hour
COP	Heat delivered / electricity consumed	Direct electric heat uses $\text{COP} \approx 1$; a heat pump requires its own temperature-dependent performance data
CAPEX / OPEX	Initial investment / recurring operating expense	Installation / annual maintenance
RC	Resistance and capacitance analogy for heat loss and heat storage	A house loses heat but also stores it in its structure
MPC	Model predictive control: plan ahead, apply the first action, measure and plan again	The prototype computes a plan; an automatic repeated control loop is future work

With one-hour intervals, 1 kWh imported in an hour means 1 kW average import during that hour. It says nothing about a kettle's short spike. Hourly peak estimates cannot size a fuse or guarantee an instantaneous grid limit.

4 - Solar production

The historical simulator uses the supplied radiation to distribute an explicitly assumed annual yield over time. Let $r[t]$ be nonnegative radiation, P the selected kWp, Y annual kWh/kWp and n the number of full years:

$$\begin{aligned} w[t] &= \max(0, r[t]) / \sum(\max(0, r)) \\ PV[t] &= P \times Y \times n \times w[t] \\ \text{Annual mean production} &= \sum(PV) / n = P \times Y \end{aligned}$$

The default **$Y = 1,000 \text{ kWh/kWp/year}$** is our editable scenario assumption. It is not derived from the irradiance file or mandated by the organiser. The yearly total is fixed by that input, while weather gives the seasonal and hourly pattern. Historical PV is therefore a planning proxy, not a calibrated panel model.

$4 \text{ kWp} \times 1,000 \text{ kWh/kWp/year} = 4,000 \text{ kWh/year}$. Moving to 6 kWp gives 6,000 kWh/year under the same yield assumption; savings do not grow proportionally because more of the extra energy may be exported at a lower price.

Roof tilt, azimuth, shading, temperature derating, inverter clipping and detailed system losses are not separately calculated in this proxy. A better input is an hourly roof-specific production series from [European Commission JRC PVGIS](#). The annual-yield control lets users explore uncertainty until these inputs are available.

An imported issued forecast may include **$p_v_kwh_per_kwp$** . Then production is that series times capacity; the annual-yield slider does not rescale a provider forecast. Without it, the issued-forecast fallback is **$radiation_wm2 / 1000 \times 0.85 \text{ kWh/kWp}$** per hour at the reference yield. The 0.85 factor is an illustrative combined-loss assumption, not a fitted parameter. Check whether a provider supplies irradiance or energy before converting.

5 - Energy balance

For each physical hour, with household demand $L[t]$ and production $PV[t]$:

```
Self[t] = min(L[t], PV[t])
Import[t] = max(0, L[t] - PV[t])
Export[t] = max(0, PV[t] - L[t])
L[t] = Self[t] + Import[t]
PV[t] = Self[t] + Export[t]
Solar self-consumption = sum(Self) / sum(PV)
Household demand covered = sum(Self) / sum(L)
```

The denominators are different. A small installation can use almost all its solar at home while covering only a small fraction of household demand. No battery is included. Daytime surplus cannot supply the evening except through a later grid purchase, accounted for separately.

For $L = 1$ kWh and $PV = 1.5$ kWh: the home uses 1 kWh of solar, imports zero and exports 0.5 kWh. At €0.08/kWh export value, it receives €0.04 of modelled credit. This is in addition to the avoided purchase of 1 kWh.

Hourly matching can overestimate immediate self-use compared with minute or meter-settlement data. Increasing temporal resolution is an accuracy improvement; it is not a reason to silently reinterpret hourly averages as simultaneous measurements.

6 - Bills, savings and simple payback

Local purchase prices in Hackathon mode are €0.18/kWh from 00-06, €0.28 from 06-17, €0.40 from 17-22 and €0.28 from 22-24. Export is €0.08/kWh. Installation is €1,300/kWp and annual maintenance is 1% of initial installation cost. These values come from [docs/HackoWatt-Common-Challenge-Assumptions.pdf](#), pages 1-2.

```

Bill = sum(Import[t] × buy[t] - Export[t] × sell[t])
Original no-PV bill = sum(original_load[t] × buy[t])
Installation cost = P × cost_per_kWp
Annual maintenance = installation_cost × maintenance_fraction
Saving A = (no-PV bill - bill_with_PV_original_habits) / n - maintenance
Saving B = (no-PV bill - bill_with_PV_shifted_activities) / n - maintenance
Simple payback A/B = net_initial_cost / annual_saving_A/B

```

Both packages use the **same original household without PV** as the baseline. B includes savings from scheduling even when those savings would also be possible without panels. It is package payback, not the isolated incremental return on PV after independently adopting a scheduler. Software, smart plugs and installation of control equipment are not included in capital cost.

If annual saving is nonpositive, payback is “Not reached.” At 0 kWp there is no PV payback, even if activity shifting saves money. Daily bill cards compare before/after activity timing with the *same* panel size; they exclude maintenance and are not the same quantity as annual investment savings.

A 1 kWh appliance moved from an evening with no PV to an hour with at least 1 kWh of unused PV avoids €0.40 of import but sacrifices €0.08 of export. Net benefit = €0.32. Describing that solar energy as costless would overstate the benefit by €0.08.

Poland mode uses PLN planning inputs and the same simplified balance. Real Polish net billing includes a deposit and settlement rules; a constant export price is not a complete bill model. The [Ministry's prosumer explanation](#) describes market-price settlement and the deposit uplift. Eligibility, settlement timing, refund limits, taxes, distribution charges and fixed fees are not implemented here. No subsidy or tax relief is granted automatically.

7 - Long-term investment

Advanced controls enable sensitivity scenarios rather than one unexplained payback number. Percentages in the interface become fractions in these equations:

```

Net initial cost = (CAPEX - min(grant, CAPEX)) × (1 - effective_tax_benefit)
PV_y[t] = PV_year1[t] × (1 - degradation)^(y-1)
Buy_y[t] = Buy_year1[t] × (1 + growth)^(y-1)
Sell_y = Sell_year1 × (1 + growth)^(y-1)
OPEX_y = OPEX_year1 × (1 + growth)^(y-1)
Replacement_y = CAPEX × replacement_fraction, only in the chosen year
Cashflow_y = avoided_bill_y - OPEX_y - Replacement_y
NPV = -net_initial_cost + sum(Cashflow_y / (1 + discount_rate)^y)

```

The chart is cumulative **discounted** cash flow. Future production is degraded and hourly balances recalculated; we do not simply reduce the entire saving by a constant percentage. The first-year shifted activity schedule is held fixed in later years. Prices, export value and maintenance share one growth input; replacement cost remains the entered share of initial CAPEX.

Hackathon defaults disable degradation, discounting, tariff growth and inverter replacement cost to preserve the common simple comparison. Poland planning defaults use 0.5% degradation, 5% discounting, 2% growth, replacement in year 15 at 15% of CAPEX and 25 years. These are editable sensitivity assumptions, not predictions of markets or warranties.

Use an installer quote for CAPEX, equipment warranty and maintenance documents for degradation/replacement, the supplier contract for tariffs, and official eligibility rules for grants/tax treatment. The effective-tax-benefit field is a simplified net-investment reduction; it does not implement a tax return or the timing of deductions.

8 - Learning habits and presence

“Learned defaults” means a transparent statistical summary. For each activity category, the builder calculates the 10th, 50th and 90th percentiles of local start time and median duration:

```

Earliest suggested start = max(0, floor(start_time_p10))
Latest suggested finish = min(24, ceil(start_time_p90 + median_duration))
Typical start = median(start_time)
Presence[kind, hour] = mean(historical_occupancy for that weekday/weekend hour)

```

The user can change these defaults. Maximum allowed movement is a design choice: six hours for laundry/dishes, one hour for cooking, two hours for TV/computer. Only laundry and dishes are enabled by default. Permission is not inferred from observed behaviour: someone using an oven at 18:00 has not consented to dinner at 15:00.

Presence is rounded to an integer for the editing interface and capped by household size. Weekday and weekend profiles provide 24 editable hours and “home all day” / “away 09-17” shortcuts. Changing them affects allowed human-activity moves and thermal gains/comfort targets. It does **not** regenerate past electricity use, hot-water demand, individual calendars or new residents' appliance use.

The replay learns from the entire supplied history, which is appropriate for retrospective exploration. A future deployed recommender must learn only from pre-issue history and confirmed future plans. A whole-house meter alone does not identify every appliance or person reliably: use appliance events, plugs, user input, or a separately validated disaggregation model.

Resident minutes and appliance minutes have different meanings. Four people watching one television for 40 minutes give 160 person-minutes but only 40 device-minutes. Device schedules reserve a physical resource once. TV and console sessions that overlap are kept at their original time to preserve their joint use.

9 - How appliance recommendations are calculated

- 1 Reconstruct each session's hourly energy from its minute profile. Laundry and dishwasher cycles retain the generator's shaped power curve; other events retain their event energy and duration.
- 2 Reserve all original sessions on each physical appliance, including devices the user does not permit to move.
- 3 For each permitted event, temporarily remove its energy from the original hours.
- 4 Try whole-hour shifts inside the user's range, keeping the original start minute, duration and local start date. Reject candidates outside the allowed start/finish window, with a physical-device overlap, or without presence for a human activity.
- 5 Calculate the marginal cost or grid objective after adding the same cycle at each candidate time. Account for lost export revenue when the cycle consumes surplus solar.
- 6 Keep the best improving candidate. Ties prefer staying closer to the original time. Reserve the selected interval and continue to the next event.

Bill goal: minimise bill change
 Solar goal: minimise $1000 \times \text{imported_kWh_change} + \text{bill_change}$
 Grid goal: minimise $1000 \times \text{change in sum}(\max(0, \text{hourly_import} - \text{grid_limit})^2) + \text{bill_change}$

The factor 1000 is an optimisation preference weight, not a physical or economic constant. It gives priority to solar use or limiting excess grid load, with money secondary. It is an approximate weighted objective, not a formal lexicographic or globally optimal solution.

The search is greedy and order-dependent. "Recommended" means the best improving time among the candidates checked for that event. It does not prove the best possible joint household schedule. Finite windows explain why the result is not always exactly noon or the cheapest tariff hour. Original start minutes, such as 15:43, are preserved from the data and are not minute-level forecast precision.

"Keep original time" locks an event and recalculates the remaining plan. If no better feasible move exists, leaving the session unchanged is a valid result. Original activities are preserved even if a newly edited permission window excludes them; that window restricts new suggestions. Existing conflicts are not silently repaired by deleting activity. Cooking-device dependency chains beyond physical resource conflicts are not modelled, so these activities remain opt-in suggestions.

Appliance timing conserves energy. Heating and hot-water energy are never treated as moveable washing-machine cycles. Thermal changes require a state model and are excluded from annual A/B scheduling savings.

10 - House physics and resolved_house.json

The generator resolves configurable distributions once, writes the resulting house parameters, then simulates that house. **resolved_house.json** is its reproducible physical description. It is not a machine-learning result or an efficiency certificate. Appliance A++/A labels do not appear as an inferred property: measured cycle kWh and operating power are the relevant inputs, and label scales/test procedures differ between products.

Field group	Meaning	Current sample
Area, height	Geometry and air volume	100 m ² ; 2.6 m
Fabric loss	Effective heat loss per floor area per degree	0.5299 W/(m ² -K)
Air changes	Baseline air replacement, plus wind/window rules	0.2019 per hour
Thermal capacity	Effective building heat storage	70.9595 Wh/(m ² -K)
Effective solar area	Combined window area/transmission proxy	6.474 m ²
Heating capacity / efficiency	Maximum electric power / conversion to heat	3.5 kW / 1.0
Tank volume / boiler power	Water heat storage / heater limit	184.235 L / 2 kW
Tank loss / setpoint	Loss conductance / thermostat target	1.6 W/K / 55 °C

The implemented house is one thermal zone with one temperature state (1R1C). For constant inputs during a short step:

```

C = floor_area × thermal_capacity_per_area / 1000 [kWh/K]
H = (floor_area × fabric_loss + 0.33 × volume × ACH) / 1000 [kW/K]
Q = heater_power × efficiency + solar_gain + internal_gain [kW]
C × dT/dt = Q - H × (T - T_out)
a = exp(-H × dt / C)
T_next = T_out + (T - T_out) × a + (Q / H) × (1 - a)

```

This exponential update is the analytic solution of our first-order heat-balance equation. It avoids a coarse explicit-Euler heating jump. It is exact only for the coefficients held constant during the step; weather, ventilation and heater actions are approximated as constant within that substep.

ACH includes infiltration, wind dependence and configured passive ventilation. The 0.33 conversion is approximately the volumetric heat capacity of air in Wh/(m³-K). Solar gain uses effective solar area × radiation; May-September gain is multiplied by 0.35 to represent the configured summer-shading assumption. Internal gains include occupants, a configured share of nonthermal electricity, and heat lost from the tank into the room.

A more detailed 2R2C model would give indoor air and the building fabric separate temperatures. It requires parameter identification and validation; it is a proposed upgrade, not an existing capability of this dashboard.

11 - Hot-water model

A well-mixed tank assumes one uniform water temperature. Its energy inventory changes with electrical heating, heat loss to the room and hot-water draw. This model family and analytic heat-balance solution are documented in the [EnergyPlus Engineering Reference, Water Thermal Tanks](#). We implement our own simplified tank model; EnergyPlus is not executed.

```
c_water ≈ 4.186 kJ/(kg-K) / 3600 ≈ 0.001163 kWh/(L-K)
C_tank = tank_volume_l × 0.001163 [kWh/K]
Q_draw = litres_delivered × 0.001163 × max(0, T_use - T_mains)
Loss = tank_loss_W_per_K / 1000 × (T_tank - T_room) [kW]
C_tank × dT_tank/dt = boiler_power - Loss - draw_heat_rate
```

In this implementation each hourly draw is split into four equal substep draws. Available energy above the requested use temperature is removed first. Once that reserve is depleted, cold-water mixing is approximated with an exponential volume exchange; unserved hot-water energy is recorded, rather than creating heat. Heating and standing losses are then integrated analytically over the 15-minute substep. Tank losses enter the room balance.

```
T_mains = 10 + 5 × sin(2π × (day_of_year - 120) / 365.25)
Usable reserve = max(0, (T_tank - T_use) × C_tank)
After reserve depletion:
T_after_draw = T_mains + (min(T_tank, T_use) - T_mains)
× exp(-remaining_draw_litres / tank_volume_litres)
```

The seasonal mains-temperature sine is an explicit approximation inherited from the simulator, not a local measurement. A 184.235 L tank stores about 0.2143 kWh per degree; heating it by 10 °C needs roughly 2.143 kWh before losses.

The current one-node model does not capture a hot top layer and cold bottom layer. [Experimental research on stratified electric-water-heater MPC](#) motivates a richer multi-node model where measured tank temperatures are available. The prototype's 50-60 °C service band is a simulation constraint, not a complete hygiene or equipment-safety controller.

12 - Thermal optimisation

The comfort view carries room and tank temperature through a 24-hour forecast. It compares a thermostat against a bounded predictive search. Both use the same house, initial state, weather, draw profile, solar and nonthermal demand.

```
Nonthermal demand = max(0, total_forecast - heating_component - water_component)
For each hour, test heating and boiler duty fractions from {0, 0.5, 1}
Simulate each candidate in four 15-minute substeps
Reject candidates with a checked temperature or hot-water-service violation
Keep up to 120 distinct low-objective states per hour
Require terminal room and tank temperatures ≥ thermostat endpoints - 0.05 °C
```

The 0.5 action represents average duty over the hour; an actual controller would need permitted switching intervals. Search-state deduplication rounds room temperature to 0.1 °C and tank temperature to 0.25 °C. The beam width and action discretisation bound computation; failure to find a plan does not prove none exists.

Minimum room targets depend on the chosen preset and presence. “Balanced” uses 20.5 °C at home, 18.5 °C during 23-06 when home, and 17.5 °C away. “Comfort first” uses 21/19/18.5 °C, and “Lower room targets” uses 20/18/17 °C. Sleeping is a clock-based approximation, not individual sleep prediction. Maximum room temperature and initial temperatures are editable.

Violations are rejected, not traded against cheap electricity. Tank temperature is checked after draw as well as after each substep; unmet hot-water demand invalidates the candidate. A terminal reserve check prevents an apparent saving from simply ending with a depleted house or tank. If the baseline is feasible it is also a candidate, preventing a worse bill under the bill objective.

If no feasible plan is found, the application displays a thermostat fallback and explicitly makes no comfort/saving claim. Initial temperatures may already violate a target; heating power is limited, and there is no active cooling. Do not lower limits simply to produce a pleasing demo result.

The default forecast's direct total and modular components are separate model estimates. Subtracting modular heat from the direct total creates a **hybrid planning profile**, not a guaranteed decomposition; negative residual demand is clamped to zero. This is disclosed and must be resolved with consistent component forecasts before real control.

Default starting temperatures are approximate previous-hour simulated means, not measured instantaneous states. The minute-resolution Python simulator remains a separate validation reference. A real MPC loop must re-read sensor state, update forecasts and execute only the first action. This static HTML does not run that loop. Thermal savings are not extrapolated to annual PV payback.

13 - Forecasts, validation and uncertainty

The repository supplies a seasonal-naive baseline, a direct total-demand gradient-boosted model and a modular component model. See [docs/forecast_models.md](#) for training features and evaluation design. The dashboard currently selects the non-naive model with the lowest recorded 24-hour WAPE; this retrospective selection is not an independent test of the selected model.

```
MAE = mean(abs(actual - prediction)) [kWh per hour]
WAPE = 100 × sum(abs(actual - prediction)) / sum(actual)
Empirical lower/upper = prediction + residual.quantile(0.10 / 0.90)
Clip negative energy limits to zero
```

The rolling evaluation trains on earlier observations and predicts later ones, following the temporal separation described in [Forecasting: Principles and Practice, time-series cross-validation](#). No future measured load, appliance demand, occupancy or thermal state may be a forecast feature. Recorded weather is nevertheless used as a perfect-weather forecast assumption in the saved demo evaluation; operational weather uncertainty will make this task harder.

Displayed uncertainty uses only residuals from earlier origins whose target observations predate the selected issue time. The range is empirical and uncalibrated, with a small number of origins. It is not a guaranteed 80% coverage interval. If no provider interval is supplied, imported lower/upper values equal the point prediction and are labelled “not provided.”

The current direct model has 24-hour MAE about 0.742 kWh and WAPE about 79.9%, over 12 forecast origins. This is a substantial error, visibly labelled in the app; do not call it production-ready prediction. Longer-horizon buckets are 25-72 and 73-168 hours, not separate cumulative 72/168-hour metrics. The table also includes the full 1-168-hour result.

Predicted peak explanations rank component *estimates*. They are not causal proof and the component sums need not equal the direct total. The forecast view itself shows unoptimised demand; replay recommendations should not be presented as tomorrow's confirmed activity plan.

14 - Integrating team models and fresh data

The current import path is a build-time contract. A new file requires rebuilding and reopening/reloading the HTML. There is no automatic refresh, streaming connection or notification service.

```
python main.py renewable-dashboard --issued-forecast results/team/issued_forecast.csv --sensor-state results/team/sensor_state.json
python main.py renewable-dashboard --model-profile results/team/model_profile.csv
```

Issued forecast field	Contract
timestamp_utc	Required; at least 24 unique, ordered, consecutive UTC hours
load_kwh	Required; nonnegative finite household energy per hour
outdoor_c	Required; finite °C, may be negative
wind_ms, radiation_wm2	Required; finite, nonnegative m/s and W/m²
issued_at_utc	Optional; one valid issue time no later than the first target. If absent, issue time is unknown
model_id	Optional model/provenance label
lower_kwh, upper_kwh	Optional nonnegative interval; lower must not exceed upper
pv_kwh_per_kwp	Optional provider production per hour per installed kWp; overrides radiation proxy
space_heating_kwh, water_heating_kwh, hot_water_draw_l	All three required to enable thermal optimisation; use consistent component forecasts
base_kwh, behaviour_kwh	Optional diagnostic components

Sensor JSON accepts `{"indoor_c":20.7,"tank_c":55}`. The supported import ranges are room 0-50 °C and tank 0-95 °C; the UI offers a narrower operating range. Sensor timestamps and freshness are not enforced by this prototype. Do that in the service before enabling control.

A replacement history CSV requires **timestamp_utc** and **load_kwh**, with exactly the same timeline as **hourly.csv**. Optional **occupancy_people** stays between 0 and 4 for this four-person generated scenario. The replacement must still support the existing appliance events; otherwise reconstruction fails. Do not feed an already shifted load back into this route and then shift the original events again.

Use [Open-Meteo's documented forecast fields](#) or another weather provider, retaining issue time, units and location. Use [Forecast.Solar](#) or an equivalent production provider for the roof-specific forecast. API access, free-tier limits and commercial terms must be checked when connecting a real service; no API credentials belong in the offline HTML.

15 - Where the constants come from

Value	Used for	Origin / interpretation
€1,300/kWp	Installation CAPEX	Common Challenge Assumptions; includes the challenge's installation package
€0.08/kWh	Export credit	Common Challenge Assumptions; not efficiency or degradation
1% / year	Maintenance	Common Challenge Assumptions
€0.18 / 0.28 / 0.40 / 0.28	Purchase time bands	Common Challenge Assumptions, Warsaw local hours
1,000 kWh/kWp/year	Historical annual PV yield	Editable team assumption; replace with roof-specific evidence
0.85	Issued-forecast irradiance fallback	Team combined-loss proxy; not used to define historical annual output
0.001163	Water heat capacity in kWh/(L-K)	Physical unit conversion, density approximated as 1 kg/L
0.33	Ventilation heat loss conversion	Approximate volumetric heat capacity of air in Wh/(m ³ -K)
0.35, May-September	Window solar gain	Generator summer-shading assumption
10 ± 5 °C, phase day 120	Seasonal mains temperature	Generator approximation, not measurements
1000	Solar/grid objective priority	Algorithm design weight
120 states, 0/50/100% duty, 15 min	Thermal search effort and resolution	Prototype design choices; no optimality guarantee
PLN 5,000/kWp; 1.10 import; 0.35 export	Poland planning mode	Illustrative editable inputs, not verified current market averages
0.5%, 5%, 2%, year 15 / 15%	Poland lifetime sensitivities	Illustrative degradation, discount, growth and replacement assumptions

Other physical and behaviour values are traceable through **run_config.json**, **resolved_house.json** and **docs/rules_v3.md**. A plausible assumption and an observed parameter are different evidence levels; show that distinction in the presentation.

16 - Worked result and verification

Reference result for the generated 2024-2025 history, 4 kWp, yield 1,000, default learned device permissions, bill objective and unmodified Hackathon economics. Values are rounded for presentation; regenerate if inputs change.

Quantity	Result	Derivation
Annual household demand	6,499.48 kWh	Sum of hourly load / 2
PV production	4,000.00 kWh/year	4 × 1,000
Solar used, current habits	1,267.00 kWh/year	Sum of hourly minima / 2
Grid imports / exports	5,232.48 / 2,733.00 kWh/year	Demand minus self-use / production minus self-use
Demand coverage / solar self-consumption	19.49% / 31.68%	1,267 / 6,499.48 and 1,267 / 4,000
Initial investment / maintenance	€5,200 / €52 per year	4 × 1,300; 1% × 5,200
Annual saving A / B	€553.16 / €589.60	Hourly bill differences minus maintenance
Simple payback A / B	9.40 / 8.82 years	5,200 divided by saving
Moveable event records / changed sessions	7,494 / 439 across two years	Encoded events / selected improving moves

The default 1 June 2025 example moves one washing cycle from 19:43 to 15:43, reducing the modelled day bill by about €0.20. There is no reason to promise large savings on every day; the purpose is to show one useful, explainable action.

In the default thermal example, thermostat cost is about €9.42 and the selected plan about €7.34. Grid energy is about 30.83 versus 31.24 kWh: **the cheaper plan uses slightly more electricity**, while improving modelled service constraints. Price timing and thermal losses explain why money and energy are different objectives. The thermostat violates the selected service limits in five hourly records; the chosen plan has none at the checked steps. This is one synthetic forecast experiment, not annual or measured field performance.

Verification checks include energy conservation, import/self-use/export identities, event-lock behaviour, non-worsening bill objective, nonnegative loads, chronological contracts and thermal bounds/end-state reserve. The repository's retained small model/contract tests do not establish real-world calibration or forecast quality.

```
python -m unittest discover -s tests -v
python main.py --help
python main.py renewable-dashboard
git diff --check
```

17 - Usability and accessibility

The interface answers one practical question per view: what can I change today; what size solar makes sense; what does my household need; why should I trust this number? Shared capacity and economic mode stay in the same place. Before/after values are adjacent, recommendations state a time and a reason, and advanced parameters are progressively disclosed.

Keyboard-operable native controls, visible focus, labelled inputs, text alongside colour, dashed comparison lines, chart data tables, larger text, higher contrast, a skip link and reduced-motion support follow the principles in [W3C's WCAG 2.2 guidance](#). Responsive layouts support narrow screens. Accessibility preferences can be remembered locally; saved scenario comparisons last for the current session.

This is not a claim of full WCAG certification. Screen-reader testing with users, touch testing, localisation, colour-vision review and usability sessions with older/less technical users remain necessary. An understandable fallback is part of usability: “no useful move found” or “forecast error is high” is more helpful than an unexplained green success badge.

18 - Five-minute demo and jury questions

Time	Show	Say
0:00-0:40	Energy plan, default June day	"A family needs a useful decision, not another complicated chart."
0:40-1:30	Before/after graph, washing recommendation, Keep original time	"One complete cycle moves inside the family's permissions; locking it updates the comparison."
1:30-2:20	Solar investment; 2, 4, 6 kWp	"These are the five required outcomes. Self-use and demand coverage answer different questions."
2:20-3:10	Presence and device insight	"History proposes defaults. The family decides what is acceptable."
3:10-4:00	Thermal comparison and temperatures	"Heat has memory. We simulate temperatures instead of sliding heating energy to noon."
4:00-4:35	Forecast, error label and range	"Forecast uncertainty is visible. Our current model needs improvement, and the team can replace it through a clear contract."
4:35-5:00	How it works and this handbook	"Every number has an input, equation and limitation. The family keeps control."

Where did the installation price and 0.08 come from?

The organiser's common-assumptions PDF. €0.08 is export value per kWh. Installation is €1,300 per kWp; neither value was learned from the data.

Is the recommendation globally optimal?

No. Appliance timing uses a constrained greedy search; thermal control uses a bounded beam search. Both are transparent and fast enough for interactive exploration. A joint mixed-integer or calibrated MPC controller is a possible next implementation.

Does this really save electricity?

Moving an appliance preserves its consumption. It can reduce cost or grid imports by matching solar. Thermal control can change consumption through losses and targets, so we report energy and money separately.

Can one TV exceed 60 minutes in an hour?

A physical TV cannot. Several residents' viewing minutes can add to more than 60; they are not added as separate televisions. Resource intervals prevent multiple concurrent sessions on the same device.

Why not say this is live AI?

The default is an honest synthetic-history and recorded-forecast demonstration. Imported issued forecasts can replace the snapshot; continuous acquisition, sensor freshness and control need a service layer.

How do you stand out?

Demonstrate reversible decisions, reasons for every move, comfort constraints, accessible controls and evidence traceability. Present the measured differences on screen rather than claiming unverified superiority over other teams.

19 - From prototype to a useful real product

- 1 **Calibrate one real home:** import meter history, installer/roof data and appliance ratings; capture room/tank sensor state with timestamps. Estimate heat-loss and heat-capacity parameters from suitable observations and quantify their uncertainty.
- 2 **Improve forecast evidence:** archive actual issued weather forecasts, use more seasonal origins, tune only on earlier data, keep a final untouched holdout and calibrate interval coverage.
- 3 **Use consistent demand components:** make their sum agree with the forecast total before thermal planning. Import predicted hot-water draw and actual future activity permissions rather than future measured behaviour.
- 4 **Add a service layer:** ingest new data, detect stale/missing sensors, refresh forecasts, replan and log decisions. Handle local daylight-saving time at the boundary while retaining UTC internally.
- 5 **Validate control:** compare the quarter-hour controller against the minute simulator, then supervised physical trials. Model switching limits, hygiene cycles, stratification, realistic roof production and uncertainty; keep manual override and equipment safety authoritative.
- 6 **Implement real Polish billing:** dated supplier tariff, distribution/fixed charges, settlement ledger, eligibility and legally applicable grant/tax rules. Display editable assumptions separately from verified contract data.
- 7 **Learn from users:** ask which recommendations they keep, measure comfort complaints and actual savings, and simplify screens based on observed difficulties.

Do not add a battery, heat pump or active cooling silently: the supplied scenario has direct electric heating and a separate electric boiler. Each additional technology needs its own physical model, cost and explicit choice.

20 - Code and reading map

Implementation	Responsibility
<code>src/hackowatt/renewable.py</code>	Tariffs, physical energy balance, historical PV shape, event encoding, habit windows and device insight payload
<code>src/hackowatt/renewable_engine.js</code>	schedule , investment , lifetime , thermal and analytic rc update
<code>src/hackowatt/components/renewable_energy.py</code>	CSV/JSON contracts, forecast provenance/intervals, offline HTML builder
<code>src/hackowatt/renewable_ui.js</code>	Editable assumptions, page state, SVG charts, comparisons and explanations
<code>src/hackowatt/renewable_dashboard.html</code> , <code>renewable_style.css</code>	Semantic interface and responsive visual design
<code>src/hackowatt/thermal.py</code>	Minute-resolution generator thermal reference
<code>docs/rules_v3.md</code> , <code>hourly_data_dictionary.md</code> , <code>forecast_models.md</code>	Generation assumptions, schema and forecasting architecture
<code>docs/thermal_mpc_design.md</code>	Current thermal implementation versus proposed richer models

Scientific references support model families and validation principles. Team assumptions and algorithm design choices are identified separately throughout this document. A cited heat-balance reference does not certify our parameter choices, forecast accuracy or savings.

Ready for the jury

The strongest closing message is simple: the family sees what changes, why it changes and which assumptions support the number. The prototype keeps user choice, comfort and uncertainty visible.

DEMO	EVIDENCE	NEXT STEP
Energy plan - Solar investment - Home & comfort - How it works	This handbook, source-backed calculations and reproducible tests	Calibrate one real home and connect issued weather/sensor data